

Amirmohammad Mohammadi

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EDUCATION

Texas A&M University, College Station, Texas	May 2027 (anticipated)
Doctor of Philosophy in Computer Engineering	
Thesis: Distribution-guided Parameter-efficient Fine-tuning of Foundation Models for Waveform-based Sensing	
Sharif University of Technology, Tehran, Iran	February 2021
Master of Science in Electrical Engineering	
Thesis: Design and Evaluation of an Integrated Biomedical System for Human Stress Detection and Monitoring	
University of Tabriz, Tabriz, Iran	September 2018
Bachelor of Science in Electrical Engineering	
Final project: Development of a Data Logger for Sensors of Smartphones	

PROFESSIONAL EXPERIENCE

Texas A&M University, College Station, Texas	September 2022 – Present
Graduate Research Assistant	
- Built a few-shot whole slide image classification pipeline with explainable AI capability to investigate parameter-efficient fine-tuning of vision-language models for alcoholic liver disease diagnosis. - Proposed a neighborhood feature pooling layer that improved remote sensing image classification accuracy by up to 2.5 percentage points over standard pooling with near-zero added overhead. - Reduced the fine-tuning parameters of Transformer-based foundation models by over 10% compared to conventional adapters by developing a distribution-aware adaptation algorithm. - Raised the classification accuracy of a convolutional deep learning model by 7 percentage points by designing a time-frequency feature engineering pipeline for audio applications. - Established a cross-domain transfer learning benchmark showing that ImageNet-pretrained models outperform audio-pretrained models by 3 percentage points in acoustic classification. - Reduced ground truth data requirements by a factor of 15 for physiological time-series signals by developing physics-informed neural networks for cuffless blood pressure measurement. - Disseminated research findings through publications, presentations, and reports to funding agencies.	
Sharif University of Technology, Tehran, Iran	July 2019 – February 2021
Graduate Student Researcher	
- Designed an ECG + EDA electrical circuit for a wearable sensor (BLE SoC) with a 3× longer battery life compared to alternatives and delivered 94% mental stress detection accuracy across 18 participants.	

SKILLS

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- **Tools:** Python; PyTorch; scikit-learn; NumPy; Pandas; Matplotlib; MATLAB; C; Hugging Face; Git; Linux; VS Code; Jupyter Notebook; High-performance Computing.
 - **Expertise:** Applied AI; Foundation Models; Deep Learning Architectures & Techniques; Generative AI; Physics-Informed Machine Learning; Multi-Modal Learning; AI Hardware; Efficient AI.

AWARDS

- Fully funded research assistantship by Massachusetts Institute of Technology (MIT) Lincoln Laboratory to advance parameter-efficient fine-tuning of foundation models (\$152,000, 2024 - 2026).
- Travel grant by IEEE Geoscience and Remote Sensing to present at conference (\$500, 2026).
- Travel grant by TAMU Electrical & Computer Engineering to present at conference (\$1000, 2026).
- Fully funded research assistantship by National Institutes of Health (NIH) to investigate physics-informed neural networks for improved cuffless blood pressure estimation (\$79,000, 2022 - 2023).
- Full tuition waivers for B.Sc. and M.Sc. electrical engineering programs based on top-tier performance in the Iran Nationwide University Entrance Exam (2014 - 2018 & 2018 - 2021).

PEER-REVIEWED PUBLICATIONS

- Neighborhood Feature Pooling for Remote Sensing Image Classification., *Orvati Nia, F., Mohammadi, A., Al Kharsa, S., Naikare, P., Hampel-Aria, Z., & Peeples, J.*, (2026). Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision. [\[link\]](#)
- Histogram-based Parameter-efficient Tuning for Passive Sonar Classification., *Mohammadi, A., Carreiro, D., Van Dine, A., & Peeples, J.*, (2026). IEEE International Geoscience and Remote Sensing Symposium. [\[link\]](#)
- Investigation of Time-Frequency Feature Combinations with Histogram Layer Time Delay Neural Networks., *Mohammadi, A., Masabarakiza, I., Barnes, E., Carreiro, D., Van Dine, A., & Peeples, J.*, (2025). IEEE OCEANS. [\[link\]](#)
- Cross-Domain Knowledge Transfer for Underwater Acoustic Classification Using Pre-trained Models., *Mohammadi, A., Kelhe, T., Carreiro, D., Van Dine, A., & Peeples, J.*, (2025). IEEE OCEANS. [\[link\]](#)
- Physics-informed neural networks for modeling physiological time series for cuffless blood pressure estimation., *Sel, K., Mohammadi, A., Pettigrew, R. I., & Jafari, R.*, (2023). Nature NPJ Digital Medicine, 6(1), 110. [\[link\]](#)
- An integrated human stress detection sensor using supervised algorithms., *Mohammadi, A., Fakharzadeh, M., & Baraeinejad, B.*, (2022). IEEE Sensors Journal, 22(8), 8216-8223. [\[link\]](#)

MANUSCRIPTS UNDER REVIEW

- Structural and Statistical Audio Texture Knowledge Distillation for Acoustic Classification., *Ritu, J., Mohammadi, A., Carreiro, D., Van Dine, A., & Peeples, J.*, [\[link\]](#)
- Parameter-efficient Dual-encoder Architecture with Differentiable Choquet Integral Fusion for Underwater Acoustic Classification., *Mohammadi, A., Van Dine, A., & Peeples, J.*,

PRESENTATIONS

- Neighborhood Feature Pooling for Remote Sensing Image Classification., *Orvati Nia, F., Mohammadi, A., Al Kharsa, S., Naikare, P., Hampel-Aria, Z., & Peeples, J.*, (2026). IEEE/CVF Winter Conference on Applications of Computer Vision (2026). Tuscon, AZ. (poster & oral presentation)
- Investigation of Time-Frequency Feature Combinations with Histogram Layer Time Delay Neural Networks., *Mohammadi, A., Masabarakiza, I., Barnes, E., Carreiro, D., Van Dine, A., & Peeples, J.*, (2024). ECE Spring 2024 Poster Event. College Station, TX. (poster presentation)

- Physics-Informed Neural Networks for Modeling Cardiovascular Dynamic., **Mohammadi, A.**, Sel, K., & Jafari, R. (2023). AI in Health Conference. Houston, TX. (poster presentation)

EXTRACURRICULARS

- **Reviewer** for the 2023 IEEE International Conference on Acoustics, Speech and Signal Processing.
- **Reviewer** for the Expert Systems With Applications international journal.
- **Helper** for the 2024 IEEE International Conference on Acoustics, Speech and Signal Processing.
- **Evaluated** student performance and provided academic support for an electronics course.
- **Mentored** undergraduate and graduate students in designing & developing end-to-end AI projects.
- **Volunteered** for Night at Zach event to demonstrate lab projects and answer questions from attendees.
- **Delivered** 3 Machine Learning lectures (neural networks/backpropagation) to a class of 99 students.

LANGUAGES

English (proficient); **Farsi** (native/bilingual); **Azerbaijani Turkish** (native/bilingual)